THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS FULL SYLLABUS PAPER SET 07

Full Syllabus (Chapters 1–8) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The distance between $A(-4, -3)$ and $B(4, 3)$ is: 1

(a) 9 units **(b)** 11 units **(c)** 12 units **(d)** 10 units

2. The distance of the point $(-5, 12)$ from the origin is: 1

(a) 13 units **(b)** 14 units **(c)** 5 units **(d)** 17 units

3. The coordinates of the origin are: 1

(a) $(1, 0)$ **(b)** $(0, 1)$ **(c)** $(0, 0)$ **(d)** $(1, 1)$

4. Which of the following is a linear polynomial? 1

(a) 7 **(b)** $x^3 - x$ **(c)** $x^2 + 1$ **(d)** $5x - 8$

5. The value of the linear polynomial $3x + 1$ at $x = 4$ is: 1

(a) 13 **(b)** 16 **(c)** -13 **(d)** 11

6. The decimal expansion of $\frac{13}{25}$ is: 1

(a) terminating **(b)** non-terminating non-repeating

(c) non-terminating repeating **(d)** cannot be determined

7. The block 142857 repeats in the decimal of: 1

(a) $\frac{1}{3}$ **(b)** $\frac{1}{7}$ **(c)** $\frac{1}{9}$ **(d)** $\frac{1}{11}$

8. Between any two rational numbers there are: 1

(a) no rational number **(b)** exactly one rational number

(c) infinitely many rational numbers **(d)** exactly two rational numbers

9. Expand $(3x - 4)^2$: 1

(a) $9x^2 + 24x + 16$ **(b)** $9x^2 - 16$ **(c)** $9x^2 - 12x + 16$ **(d)** $9x^2 - 24x + 16$

10. $(x - 6)(x - 3)$ equals: 1

(a) $x^2 + 18x - 9$ **(b)** $x^2 - 3x + 18$ **(c)** $x^2 - 9x - 18$ **(d)** $x^2 - 9x + 18$

11. An arc subtends an angle of 150° at the centre. The angle it subtends at a point on the remaining part of the circle is: 1

(a) 150° **(b)** 105° **(c)** 75° **(d)** 300°

12. A chord of length 12 cm is drawn in a circle of radius 10 cm. The distance of the chord from the centre is: 1

(a) 16 cm **(b)** 8 cm **(c)** 4 cm **(d)** 7 cm

13. ABCD is a cyclic quadrilateral with $\angle A = 92^\circ$. Then $\angle C$ is: 1

(a) 92° **(b)** 88° **(c)** 268° **(d)** 44°

14. By Brahmagupta's formula, the area of a cyclic quadrilateral with sides $3, 4, 3, 4$ is: 1

(a) 24 sq. units **(b)** 10 sq. units **(c)** 16 sq. units **(d)** 12 sq. units

15. The length of an arc subtending 180° at the centre of a circle of radius 14 cm ($\pi = \frac{22}{7}$) is: 1

(a) 88 cm **(b)** 44 cm **(c)** 46 cm **(d)** 22 cm

16. The area of a triangle with base 10 cm and height 7 cm is: 1

(a) 45 cm^2 **(b)** 17 cm^2 **(c)** 35 cm^2 **(d)** 70 cm^2

17. The game of Snakes and Ladders originated in ancient India from a game called: 1

(a) Shatranj **(b)** Pachisi only **(c)** Mancala **(d)** Jñān-Chaupaḍ

18. A bag contains 3 red, 9 blue and 3 green balls. One ball is drawn at random. The probability that it is blue is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{3}{5}$ **(c)** $\frac{2}{5}$ **(d)** $\frac{4}{5}$

19. For the sequence $t_n = 5n - 3$, the value of t_{10} is: 1

(a) 46 **(b)** 48 **(c)** 47 **(d)** 94

20. The 7th term of the GP with first term 1 and common ratio -2 is: 1

(a) 128 **(b)** 63 **(c)** 65 **(d)** 64

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 20–30 words each.

21. Is the point $(-5, -12)$ at the same distance from the origin as $(13, 0)$? Justify. 2

22. The sum of two numbers is 64 and one is 10 more than the other. Form a linear equation and find the numbers. 2

23. A spice trader has a loan of ₹850, makes a profit of ₹1,200 and then a loss of ₹450. Write the sum using integers and find his final standing. 2

24. Find 55^2 and 65^2 using $a^2 = (a + b)(a - b) + b^2$ with $b = 5$. 2

25. An arc subtends an angle of 50° at a point on the circle. What angle does it subtend at the centre? 2

26. Find the perimeter of a quadrant (quarter-circle) of radius 14 cm, including the two radii. ($\pi = \frac{22}{7}$) 2

27. A spinner has 8 equal sectors numbered 1 to 8. Find the probability that it stops on a multiple of 3. 2

28. Which term of the AP $21, 18, 15, \ldots$ is -81? Is 0 a term? 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each.

29. Three vertices of a rectangle are $(2, 1)$, $(8, 1)$ and $(8, 5)$. Find the fourth vertex and the length of the diagonal. 3

30. Solve the linear equation $3x - 4 = 11$ and verify the solution. Also write a word problem that leads to $2x + 10 = 64$. 3

31. A rational number in its lowest form has denominator $2^3 \times 5$. How many decimal places will its expansion have? Explain. 3

32. The breadth of a pool is 4 m less than its length and its area is 96 m². Find its dimensions. 3

33. ABCD is a parallelogram of area 84 cm^2. Find the area of $\triangle ABD$ and justify. 3

34. Explain what happens to experimental probability as the number of trials increases, using the Law of Large Numbers. 3

35. In a circle of radius 10 cm there are two chords at distances 6 cm and 8 cm from the centre. Find their lengths and show that the nearer one is longer. 3

OR

If two equal chords of a circle intersect inside the circle, show that the segments of one chord are equal to the corresponding segments of the other. 3

36. Which term of $t_n = 3n - 7$ is 332? Is 557 a term of this sequence? 3

OR

An AP has 50 terms; its 3rd term is 12 and its last term is 106. Find the 29th term. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. **Case study:** A square has side 1 unit. 5

(i) Find the length of its diagonal.

(ii) Show how to represent it on the number line.

(iii) Why can this number not be written as $\frac{p}{q}$?

(iv) Write its decimal expansion to 4 places and say what kind of decimal it is.

OR

(i) Without long division, state which of $\frac{13}{80}$, $\frac{5}{12}$, $\frac{29}{343}$ have terminating decimals. 5

(ii) For the terminating one, find the decimal expansion.

(iii) Find the decimal of $\frac{5}{12}$ up to 5 places and state the repeating digit.

38. **Case study:** A circular park has radius 25 m. Two straight paths (chords) of lengths 14 m and 48 m are laid. 5

(i) Find the distance of each path from the centre.

(ii) Which is nearer the centre and why?

(iii) Find the distance between the paths if they are parallel and on opposite sides of the centre.

(iv) Find it if they are on the same side.

OR

Prove that the sum of the opposite angles of a cyclic quadrilateral is 180°. 5

39. **Case study:** A circular park of radius 35 m has a circular fountain of radius 7 m at its centre. ($\pi = \frac{22}{7}$) 5

(i) Find the area of the park.

(ii) Find the area of the fountain.

(iii) Find the area of grass.

(iv) Find the length of railing needed to fence the fountain and the park boundary together.

OR

Two congruent circles of radius r with centres A and B each pass through the other's centre and meet at C and D. 5

(i) Prove that $\triangle ABC$ is equilateral.

(ii) Find $\angle CAD$.

(iii) Find the perimeter of the outer boundary of the figure.

40. (i) Plot the points (n, t_n) for the AP $1, 5, 9, 13, 17$ and say what you observe. 5

(ii) Plot the points for the GP $3, 6, 12, 24, 48$ and compare.

(iii) What does this tell you about growth in an AP and a GP?

OR

(i) Show that $25 + 26 + \cdots + 58 = 1411$ using $S_{58} - S_{24}$. 5

(ii) Check the answer using 'average of first and last × number of terms'.

(iii) Who first recorded this result and where?

END OF QUESTION PAPER **Total: 80 Marks**

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